

IOT Based Battery Management System

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By

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ABSTRACT

A Battery Management System (BMS) is an essential part of modern battery-powered systems, ensuring safe operation, monitoring, and protection of batteries. This project presents an IoT-based Battery Management System using ESP32 that continuously monitors battery voltage, battery percentage, and temperature in real time. The measured parameters are displayed locally on an OLED display and uploaded to the ThingSpeak cloud platform for remote monitoring.

The system uses a DS18B20 temperature sensor for temperature measurement and a potentiometer to simulate battery voltage. Safety mechanisms such as LEDs, buzzer alerts, and relay-based load disconnection are implemented to protect the battery from over-temperature and low-battery conditions. The integration of IoT technology enables remote visualization and analysis of battery parameters, making the system suitable for smart energy storage applications, electric vehicles, and renewable energy systems.

Keywords: Battery Management System, ESP32, IoT, ThingSpeak, OLED Display, Temperature Monitoring, Relay Protection.

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Introduction

The rapid growth of electric vehicles (EVs), renewable energy systems, portable electronics, and energy storage applications has increased the demand for efficient battery monitoring and protection systems. Batteries are critical energy storage devices, and their performance, lifespan, and safety depend on proper operating conditions. Excessive temperature, over-discharge, overcharging, and improper load management can significantly reduce battery life and may even lead to safety hazards.

A Battery Management System (BMS) is an electronic system designed to monitor, control, and protect batteries during operation. It continuously measures important battery parameters such as voltage, temperature, charging status, and battery health. Based on these measurements, the BMS can take preventive actions to protect the battery and connected devices.

In this project, an IoT-based Battery Management System has been developed using the ESP32 microcontroller. The system monitors battery voltage using an analog input and measures temperature using the DS18B20 digital temperature sensor. The collected data is displayed locally on an OLED display and simultaneously transmitted to the ThingSpeak cloud platform for remote monitoring and analysis.

To improve safety and reliability, the system includes multiple protection mechanisms. A relay module is used to disconnect the load whenever abnormal conditions such as low battery percentage or high temperature are detected. Visual and audible alerts are provided through LEDs and a buzzer, enabling users to quickly identify fault conditions. The OLED display provides real-time information including voltage level, battery percentage, temperature, and relay status.

The integration of IoT technology enables remote monitoring of battery parameters from anywhere through an internet connection. Data stored on the cloud can be visualized using graphs and used for future analysis, predictive maintenance, and performance evaluation.

PROBLEM STATEMENT AND OBJECTIVES

PROBLEM STATEMENT

Traditional battery systems often operate without proper monitoring and protection. Users may not receive timely warnings about overheating or low battery conditions, resulting in reduced battery life and safety risks.

Therefore, there is a need for an intelligent and low-cost Battery Management System capable of:

- Monitoring battery parameters
- Providing safety protection
- Displaying real-time information
- Sending data to the cloud

OBJECTIVES

The main objectives of the project are:

1. Measure battery voltage using ESP32 ADC.
2. Calculate battery percentage.
3. Monitor battery temperature using DS18B20.
4. Display data on OLED display.
5. Upload battery parameters to ThingSpeak cloud.
6. Provide visual and audible alerts.
7. Disconnect load during unsafe conditions using relay.
8. Improve battery safety and monitoring.

Literature Review

Battery Management Systems (BMS) have become an essential component in modern energy storage applications such as electric vehicles, renewable energy systems, portable electronic devices, and industrial backup power systems. The primary purpose of a BMS is to monitor battery parameters, ensure safe operation, improve performance, and extend battery lifespan. Over the years, researchers and engineers have developed various techniques and technologies to improve battery monitoring and protection.

Traditional battery monitoring systems were mainly limited to measuring battery voltage and providing basic protection against overcharging and deep discharge. Although these systems were simple and cost-effective, they lacked advanced monitoring capabilities and remote access features. As battery-powered applications became more complex, the need for intelligent battery management systems increased significantly.

Several studies have focused on the use of microcontrollers for battery monitoring applications. Microcontrollers such as Arduino, PIC, STM32, and ESP32 have been widely used due to their ability to process sensor data and implement protection algorithms. Among these, ESP32 has gained significant popularity because of its high processing power, low cost, and built-in Wi-Fi connectivity. These features make ESP32 highly suitable for Internet of Things (IoT) applications.

Recent developments in IoT technology have enabled battery parameters to be monitored remotely through cloud platforms. Researchers have integrated battery monitoring systems with cloud services such as ThingSpeak, Blynk, Firebase, and AWS IoT. These platforms allow users to visualize battery data in real time, store historical records, and analyze system performance through graphical dashboards. Remote monitoring helps in predictive maintenance and reduces the risk of battery failures.

Temperature monitoring has also become an important aspect of modern battery management systems. Excessive battery temperature can reduce battery efficiency, shorten battery life, and create safety hazards. Digital temperature sensors such as DS18B20 are commonly used because they provide accurate temperature measurements and easy interfacing with microcontrollers. Many studies have demonstrated that temperature-based protection mechanisms significantly improve battery reliability and safety.

Protection systems using relays, LEDs, and buzzers have been widely adopted in battery management applications. Relay modules are used to disconnect the load whenever unsafe operating conditions are detected. Visual indicators such as LEDs provide immediate status information, while buzzers generate audible alerts to notify users of fault conditions. These protection mechanisms help prevent battery damage caused by overheating or excessive discharge.

Based on the review of existing battery management systems, it is observed that the integration of ESP32, IoT platforms, temperature monitoring, and relay-based protection

HARDWARE REQUIREMENTS

ESP32

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth connectivity. It acts as the main controller of the system.

OLED Display (SSD1306)

The OLED display provides real-time visualization of battery voltage, percentage, temperature, and relay status.

DS18B20 Temperature Sensor

The DS18B20 is a digital temperature sensor used to monitor battery temperature accurately.

Relay Module

The relay disconnects the load whenever unsafe battery conditions occur.

Buzzer

The buzzer provides audible warning signals.

LEDs

- Green LED → Normal operation
- Red LED → Fault condition

Potentiometer

Used in simulation to represent battery voltage variation.

SOFTWARE REQUIREMENTS

- Arduino IDE
- Wokwi Simulator
- ThingSpeak IoT Platform
- Adafruit SSD1306 Library
- Adafruit GFX Library
- OneWire Library
- DallasTemperature Library
- WiFi Library
- HTTPClient Library

SYSTEM ARCHITECTURE

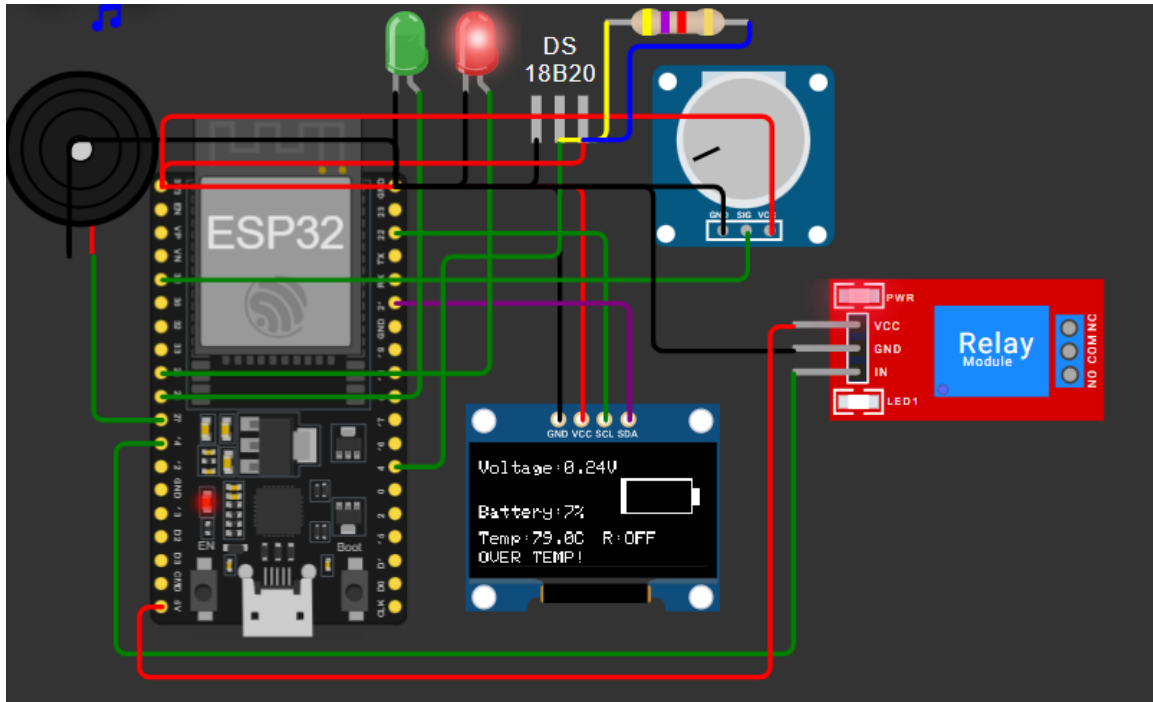


Figure: System Architecture of IoT Based Battery Management System

The system consists of ESP32 as the central controller. The OLED display shows battery parameters locally. DS18B20 measures temperature. Relay, buzzer, and LEDs provide protection functions. ThingSpeak stores and visualizes data remotely

Working Principle

The IoT-Based Battery Management System continuously monitors battery voltage and temperature using the ESP32 microcontroller. A potentiometer is used to simulate battery voltage, while the DS18B20 sensor measures temperature. The ESP32 processes these values and calculates the battery percentage.

The measured parameters are displayed on the OLED screen, allowing real-time monitoring of battery status. Simultaneously, the ESP32 sends voltage, battery percentage, and temperature data to the ThingSpeak cloud platform through Wi-Fi for remote monitoring and analysis.

The system also provides battery protection features. If the battery percentage falls below 20% or the temperature exceeds 45°C, the ESP32 activates the protection mechanism. In this condition, the red LED and buzzer are turned ON, and the relay disconnects the load to prevent battery damage. When the battery returns to safe operating conditions, the relay reconnects the load, the green LED turns ON, and normal operation resumes.

Thus, the system ensures continuous monitoring, remote accessibility, and protection of the battery, improving its safety, reliability, and lifespan.

Working Sequence

1. Read battery voltage.
2. Read battery temperature.
3. Calculate battery percentage.
4. Display data on OLED.
5. Upload data to ThingSpeak.
6. Check safety conditions.
7. If Battery < 20% or Temperature > 45°C:
 - Relay OFF
 - Red LED ON
 - Buzzer ON
8. Otherwise:
 - Relay ON
 - Green LED ON
9. Repeat the process continuously.

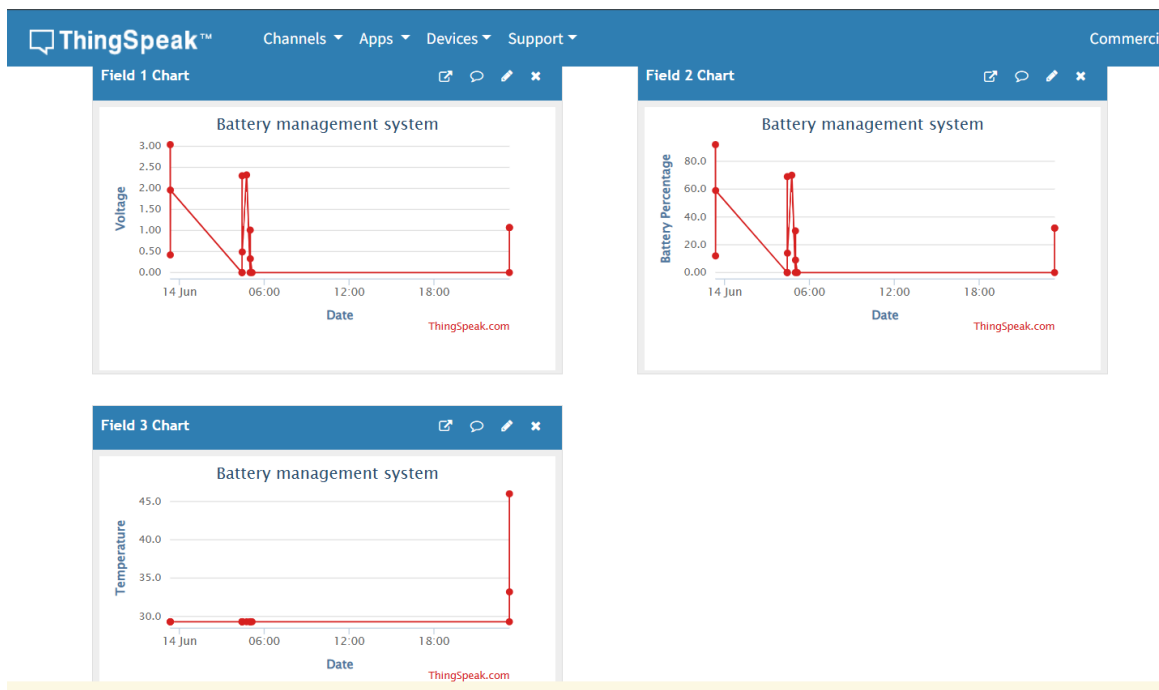
THINGSPEAK INTEGRATION

ThingSpeak is a cloud-based IoT analytics platform used for collecting, storing, and visualizing sensor data.

The ESP32 sends:

- Field 1 → Voltage
- Field 2 → Battery Percentage
- Field 3 → Temperature

The platform generates graphical representations of the collected data for analysis.



Results and Observations

11. Results and Observations

The proposed IoT-Based Battery Management System was successfully designed and implemented using the ESP32 microcontroller. The system was tested in the Wokwi simulation environment to evaluate its monitoring, protection, and cloud connectivity features. During testing, all hardware and software modules performed according to the design requirements.

The battery voltage was monitored using the analog input of the ESP32, where a potentiometer was used to simulate different battery voltage levels. The measured voltage values were accurately converted into battery percentage and displayed on the OLED screen. As the voltage was varied, the battery percentage changed accordingly, demonstrating proper system functionality.

The DS18B20 temperature sensor successfully measured and updated temperature readings in real time. Temperature values were displayed on the OLED screen and continuously monitored by the ESP32. The system responded effectively to temperature changes and accurately detected over-temperature conditions.

The OLED display provided a clear and user-friendly interface for monitoring battery parameters. It displayed battery voltage, battery percentage, temperature, relay status, and warning messages. This allowed users to observe the battery condition instantly without requiring external devices.

The IoT functionality of the system was verified using the ThingSpeak cloud platform. The ESP32 successfully connected to the Wi-Fi network and transmitted voltage, battery percentage, and temperature data to the cloud. The received data was displayed in graphical form on the ThingSpeak dashboard, enabling remote monitoring and analysis. The cloud platform also maintained historical records of battery performance, which can be useful for future analysis and maintenance purposes.

The protection mechanism was tested under various fault conditions. When the battery percentage dropped below 20%, the system activated the red LED and buzzer while disconnecting the load through the relay. Similarly, when the temperature exceeded 45°C, the relay was switched OFF automatically, preventing potential battery damage. During fault conditions, warning messages such as “LOW BATTERY” and “OVER TEMP” were displayed on the OLED screen, providing immediate user notification.

Overall, the developed system successfully achieved its objectives of real-time battery monitoring, fault detection, cloud-based data logging, and automatic protection. The experimental results demonstrated that the proposed Battery Management System is reliable, efficient, and suitable for smart battery monitoring applications. The integration

of ESP32, IoT technology, and safety features significantly improves battery management and operational safety.

Key Observations

- Real-time voltage monitoring was achieved successfully.
- Battery percentage was calculated accurately.
- Temperature monitoring functioned reliably.
- OLED display showed all parameters clearly.
- ThingSpeak received and stored data successfully.
- Relay protection worked correctly during fault conditions.
- LED and buzzer alerts provided effective warning indications.
- The overall system operated reliably and responded quickly to changing conditions.

Advantages

Advantages

The proposed IoT-Based Battery Management System offers several advantages over conventional battery monitoring methods. The system provides real-time monitoring of important battery parameters such as voltage, battery percentage, and temperature, allowing users to continuously track battery performance.

One of the major advantages of the system is its ability to provide automatic protection against unsafe operating conditions. When the battery voltage drops below a predefined limit or the temperature exceeds the safe operating range, the system automatically disconnects the load using a relay. This helps prevent battery damage and improves overall safety.

The integration of the ESP32 microcontroller provides built-in Wi-Fi connectivity, enabling seamless communication with cloud platforms such as ThingSpeak. As a result, battery data can be monitored remotely from any location with internet access. This feature is particularly useful for applications where batteries are installed in remote or difficult-to-access locations.

The OLED display provides a user-friendly interface by displaying all important battery parameters in real time. Additionally, LED indicators and buzzer alerts provide immediate visual and audible notifications whenever fault conditions occur.

The system is also cost-effective because it uses low-cost and widely available electronic components. Its modular design makes it easy to expand and customize for different battery applications.

Advantages of the System

- Real-time battery monitoring
- Automatic protection against faults
- Remote monitoring through IoT
- User-friendly OLED display
- Visual and audible fault indications
- Low-cost implementation
- Easy scalability and customization
- Improved battery safety and lifespan
- Cloud-based data storage and analysis
- Suitable for educational and industrial applications

Applications

Battery Management Systems are widely used in modern electronic and energy storage systems. The proposed IoT-Based Battery Management System can be applied in various sectors where battery monitoring and protection are essential.

One of the primary applications is in electric vehicles, where battery safety and performance play a critical role. Continuous monitoring of voltage and temperature helps prevent battery failures and improves vehicle reliability.

The system can also be used in renewable energy systems such as solar and wind energy storage units. In such systems, batteries store generated energy and require continuous monitoring to ensure efficient operation and long service life.

Industrial battery banks used for backup power applications can benefit from remote monitoring and fault detection capabilities provided by the system. Similarly, uninterruptible power supply (UPS) systems can use the proposed solution to monitor battery health and prevent unexpected failures.

Portable electronic devices, smart home energy systems, telecommunications equipment, and research laboratories can also utilize this technology for battery monitoring and analysis.

Applications of the System

- Electric Vehicles (EVs)
- Solar Energy Storage Systems
- Wind Energy Storage Systems
- Uninterruptible Power Supply (UPS)
- Industrial Battery Banks
- Telecommunications Equipment
- Portable Electronic Devices
- Smart Home Energy Systems
- Research and Educational Projects
- IoT-Based Monitoring Systems

Future Scope

Although the developed Battery Management System successfully performs monitoring and protection functions, several advanced features can be incorporated in future versions to improve system performance and functionality.

One possible enhancement is the integration of current sensors such as INA219 or ACS712. Current monitoring would allow the system to calculate battery power consumption, charging current, and overall battery efficiency more accurately.

The implementation of State of Charge (SOC) and State of Health (SOH) estimation algorithms can provide a more accurate assessment of battery condition. These techniques are commonly used in advanced Battery Management Systems for electric vehicles and energy storage applications.

A dedicated mobile application can be developed to allow users to monitor battery status directly from smartphones. Push notifications and alert messages can also be implemented to notify users whenever abnormal conditions occur.

The integration of GSM or LoRa communication modules can further extend monitoring capabilities to areas where Wi-Fi connectivity is unavailable. Artificial Intelligence (AI) and Machine Learning (ML) techniques can also be applied to predict battery failures and optimize battery performance.

Future versions may include multi-cell battery monitoring and balancing circuits, making the system suitable for large battery packs used in electric vehicles and industrial applications.

Future Enhancements

- Current monitoring using INA219 or ACS712
- Battery State of Charge (SOC) estimation
- Battery State of Health (SOH) monitoring
- Mobile application development
- GSM/SMS alert system
- LoRa-based long-distance communication
- AI-based battery health prediction
- Multi-cell battery balancing
- Data analytics and predictive maintenance
- Integration with electric vehicle battery packs

Conclusion

The IoT-Based Battery Management System developed in this project successfully demonstrates an intelligent and efficient solution for battery monitoring and protection. The system continuously monitors battery voltage, battery percentage, and temperature while providing real-time information through an OLED display.

The integration of the ESP32 microcontroller enables wireless communication and cloud connectivity through the ThingSpeak platform. This allows users to monitor battery performance remotely and analyze historical data using graphical representations. The implementation of IoT technology significantly improves accessibility and usability compared to traditional battery monitoring systems.

Safety features such as relay-based load disconnection, LED indicators, and buzzer alerts ensure reliable battery protection under low-battery and high-temperature conditions. These features help prevent battery damage, enhance operational safety, and increase battery lifespan.

The project successfully achieved all its objectives and demonstrated the practical application of IoT technology in battery management. Due to its low cost, ease of implementation, and scalability, the proposed system can be further developed for use in electric vehicles, renewable energy systems, industrial battery banks, and other smart energy applications.

In conclusion, the developed Battery Management System provides an effective combination of monitoring, protection, and cloud-based data management, making it a reliable solution for modern battery-powered systems.

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